

CSCE 650 VIRTUAL REALITY · MODULE 01 · LECTURE NOTES

Presence, and What VR Must Deliver

A correctness criterion that lives in somebody's head.

MODULE OUTCOMES

Define presence and distinguish it from immersion.

Explain why VR has a hard real-time requirement.

State the frame budget and what it is spent on.

Identify what breaks presence and what does not.

Explain why the evaluation criterion is perceptual.

1 Presence

	Immersion	Presence
Belongs to	The system.	The person.
Means	The objective extent to which the display and tracking replace real sensory input: field of view, resolution, refresh rate, tracking degrees of freedom, latency.	The subjective sense of <i>being there</i> — and, more usefully, the tendency to respond to virtual events as though they were real: flinching, stepping back, reaching out.
Established by	Measurement. A spec sheet describes immersion.	Report and behaviour. Questionnaires, and observed responses.
Relationship	Immersion <i>enables</i> presence.	It does not guarantee it. A technically excellent system with one consistent flaw produces less presence than a modest one without it.

PRESENCE IS FRAGILE, AND THE ASYMMETRY IS THE DESIGN RULE

Presence accumulates slowly. It is built from consistency — things behaving as expected, repeatedly, until the user stops checking. Minutes, typically.

It breaks in a single frame. A hand passes through a table; a footstep comes from the wrong direction; the world judders once. The user is instantly aware of the headset again.

And it does not immediately return. Having been reminded the world is synthetic, the user begins watching for the next failure, which makes the next failure easier to notice.

So the design objective is not to maximise the best moments but to eliminate the worst ones. A uniformly adequate experience sustains more presence than an excellent one with a recurring flaw — which inverts the usual instinct in graphics, where the impressive moment is what gets built.

Breaks presence	Why
Latency	The world lags the head, so it feels attached to you rather than independent of you. Module 06.
Tracking failure	Position jumps or drifts. The world moves when you did not.
Penetration	A hand or held object passes through solid geometry. The most common break by a wide margin , because users test it deliberately within the first minute.
Wrong scale	A room subtly the wrong size, or a doorway too small. Users rarely identify what is wrong, only that the space feels dreamlike.
Mismatched audio	A sound from the wrong direction, or silence where a sound was expected. Module 11.
Not on this list	Polygon count, texture resolution, shadow quality, lighting fidelity. They contribute, and they matter less than every item above. This surprises people arriving from CSCE 641, where those are the whole subject.

2 The real-time requirement

YOU ARE RENDERING TO THE VESTIBULAR SYSTEM

The vestibular system in the inner ear measures head rotation and linear acceleration **directly and continuously**, with a latency of a few milliseconds. It does not need to recognise anything or process an image; it is an accelerometer wired to the brainstem.

The display, meanwhile, reports where the head was some milliseconds ago — however many the pipeline takes.

When those two signals disagree, the brain has a conflict to resolve. One of its established responses to sustained sensory conflict is nausea, for reasons Module 07 covers.

So a late frame is not a performance complaint; it is a physiological event. No amount of visual quality compensates for it, and the usual engineering instinct — 'it runs at 45, that is playable' — is simply wrong here. **This is the single most important difference between VR and every other kind of real-time graphics.**

Refresh rate	Budget per frame	Assessment
60 Hz	16.7 ms	Below the comfortable floor for most people. Early headsets used it and the discomfort was notorious.
72 Hz	13.9 ms	The minimum on most standalone headsets. Acceptable for seated, low-motion content.
90 Hz	11.1 ms	The usual target. Where most people stop noticing the display's discreteness.
120 Hz	8.3 ms	Noticeably better for fast motion; expensive.
144 Hz	6.9 ms	Diminishing returns for most content.

And that budget covers both eyes. At 90 Hz you have 11.1 ms to produce two views, which — after the setup that can be shared between them — is on the order of 5.5 ms per eye for everything: geometry, shading, post, and submission. This is the figure students consistently underestimate, because the number quoted is per frame and the work is per eye.

HIT THE BUDGET FIRST; SPEND THE REMAINDER ON QUALITY

When a frame misses its deadline, the compositor must do something. At best it reprojects the previous frame to the current head pose (Module 06); at worst it displays the previous frame unchanged. **Either way the world judders**, and the judder is correlated with head motion, which is exactly the condition that causes discomfort.

Users reliably notice a single dropped frame. They do not reliably notice a 30% reduction in texture resolution, or one fewer shadow cascade, or a cheaper reflection.

So the ordering is fixed: meet the frame budget first, then spend whatever remains on visual quality.

In ordinary real-time graphics these can be traded against each other; here one of them is a hard constraint and the other is not.

And profile the 99th percentile rather than the mean. A mean frame time of 9 ms is consistent with one frame in fifty taking 20 ms, and those are precisely the frames that cause discomfort. **The mean is the statistic that hides the problem**, which is why Project 1 asks for a histogram.

3 What VR invalidates

Assumption from CSCE 641	In VR
One view rendered per frame.	Two , with some setup shareable and most work doubled.
The camera is authored — the developer decides what is on screen.	The user controls it absolutely. See below.
A dropped frame is a visible stutter and a minor annoyance.	A dropped frame is a physiological event. §2.
The screen is a flat surface at a fixed distance; depth is implied by cues.	Depth is genuinely perceived through stereopsis and parallax, so errors in it are directly visible rather than merely incorrect.
Post-processing effects are a matter of taste and budget.	Motion blur and heavy temporal antialiasing are actively harmful — they add perceived latency and smear during head motion, which is constant.
Field of view is a design decision.	Fixed by the hardware , and changing it is both wrong and nauseating.

YOU HAVE LOST CONTROL OF THE CAMERA

This is the design consequence with the widest reach, and it is consistently underestimated by people arriving from games or film.

Every technique that depends on framing is unavailable. No cuts, no camera moves, no composition, no control over what is on screen at any moment. The user decides, continuously.

So they will look at the thing you did not finish, stand somewhere you did not anticipate, lean through a wall to see what is behind it, and examine the back of an object you modelled from one side. Everything must be finished in every direction, and the budget has to accommodate that.

Attention must be directed by light, sound, and motion — the techniques of theatre rather than cinema. A moving object, a sound from a direction, a brightening: these draw the eye without taking control.

Taking the camera from the user to show them something is the classic beginner mistake. It is unpleasant, it breaks presence completely, and because the motion is unanticipated by the vestibular system it is a reliable way to make someone ill (Module 07).

A PERCEPTUAL CRITERION DEMANDS AN EMPIRICAL METHOD

There is no equation whose value tells you the experience is correct. CSCE 647 could difference an image against a converged reference; CSCE 649 could plot conserved energy; CSCE 608 could kill the process and check. This course has no equivalent.

The measurements that do exist are proxies. Frame time, motion-to-photon latency, tracking jitter, stereo disparity error — all necessary, none sufficient. A system can be correct on every one and still be unpleasant.

The actual criterion is whether people feel present and do not feel ill, and the only way to establish that is to ask people: several of them, including people who are not like you.

Your own tolerance is not data. People who build VR are a self-selected population with unusually high tolerance — they acclimatised early and kept going precisely because it did not bother them. **This is a large part of why so much uncomfortable VR ships,** and it is why both projects in this course require testing on people who are not you, including at least one novice.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapters 1 and 12 (free)	The definition of presence, the history, and the evaluation chapter. Chapter 12 is the empirical-method argument of §3, properly developed.
Michael Abrash — What VR Could, Should, and Almost Certainly Will Be (free talk)	The clearest statement of why the requirements are what they are, from the period when they were being discovered.
Meta — VR design guidelines and comfort documentation (free)	The practical rules, with the numbers. Read the comfort section before Module 07.
Slater — Place illusion and plausibility (free)	The research framing of presence as two separable illusions — being there, and the events being real. More precise than the usual treatment.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Put on a headset and write down, within the first two minutes, every moment you became aware of the hardware. This list is your design brief.

- 2 Find a VR application and deliberately try to break presence: reach through geometry, look behind objects, stand where you should not. Record what succeeded.
- 3 Measure your own IPD, and then try an application with it set 10 mm too wide and 10 mm too narrow. Describe each from the inside.
- 4 Take a frame-time capture from any VR application over five minutes. Plot the histogram and report the mean, the 95th, and the 99th percentile. Note how different the story is.
- 5 Compute the per-eye budget at 72, 90, and 120 Hz, and compare against a measured per-eye GPU time from a scene you build.
- 6 Show a VR application to someone who has never used one. Say nothing and record what they do in the first sixty seconds.
- 7 Find a VR application that moves the camera without user input. Describe the sensation precisely.
- 8 Write a page on an experience that produced strong presence for you, and identify which of §1's factors it got right.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Distinguish immersion from presence, and say how they are each established.
- 2 Why is presence asymmetric, and what design rule follows?
- 3 Name five things that break presence, and three things that matter less than people expect.
- 4 Why does rendering to the vestibular system change the frame requirement?
- 5 State the per-frame and per-eye budget at 90 Hz.
- 6 Why profile the 99th percentile rather than the mean?
- 7 Give five assumptions from ordinary real-time graphics that VR invalidates.
- 8 What follows from losing control of the camera, and what replaces framing?
- 9 Why must the evaluation method be empirical, and why is your own tolerance not data?